

Exercise 5 - Magnetic exchange coupling in a Cu(II) dimer from broken-symmetry DFT

The compound is a paddle-wheel dimer with four trifluoroacetate bridges between the two Cu(II) ions (Cu–Cu 2.73 Å) and an alkoxide-type O donor in each axial position; the total charge is –2 as given. Each Cu(II) is d^9 with $S = 1/2$. Keywords: ! UKS B3LYP <basis> <auxiliary basis> RIJCOSX TightSCF SlowConv, with %scf BrokenSym 1,1 end and the coordinate line * xyzfile –2 3 (high-spin triplet). ORCA first converges the high-spin state (HS, both spins parallel), then flips the spin on one Cu and converges the broken-symmetry state (BS). The three runs differ only in the relativistic treatment: non-relativistic (def2-SVP, def2/J), DKH (DKH-def2-SVP, SARC/J) and ZORA (ZORA-def2-SVP, SARC/J). The assignment does not specify the non-relativistic level, so B3LYP/def2-SVP was used to keep the functional and basis-set size the same in all three.

Results

Table 1. Total energies of the high-spin and broken-symmetry states, $\langle S^2 \rangle$ values and exchange coupling constants J in cm^{-1} . ORCA uses $H = -2J S_A \cdot S_B$, so $J < 0$ means antiferromagnetic coupling. $J(1)$: Ginsberg–Noodleman, $J(2)$: Bencini–Gatteschi, $J(3)$: Yamaguchi, $J = (E_{BS} - E_{HS}) / (\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS})$.

Run	E(HS) / E_h	E(BS) / E_h	$\langle S^2 \rangle_{HS}$	$\langle S^2 \rangle_{BS}$	E(HS)–E(BS) / cm^{-1}	J(1)	J(2)	J(3)
non-relativistic	–6076.975267	–6076.975094	2.0050	1.0006	–38.0	+38.0	+19.0	+37.8
DKH	–6107.249506	–6107.249605	2.0041	0.9989	+21.7	–21.7	–10.9	–21.6
ZORA	–6122.764932	–6122.765176	2.0038	0.9976	+53.5	–53.5	–26.8	–53.2

The broken-symmetry states are the intended ones. The Mulliken spin populations on the two Cu atoms are +0.63 and –0.63 (non-relativistic; +0.72 / –0.72 with ZORA), and $\langle S^2 \rangle_{BS} \approx 1.0$ compared with $\langle S^2 \rangle_{HS} \approx 2.0$, as expected for two localised $S = 1/2$ centres with opposite spins.

What the table shows

The relativistic treatment lowers the total energy by 30.3 E_h (DKH) and 45.8 E_h (ZORA). Almost all of this comes from the contraction of the Cu 1s–2p core shells, which is the same in both spin states and cancels in $E(HS) - E(BS)$. The absolute energies of different relativistic Hamiltonians cannot be compared with each other; only energy differences within one Hamiltonian are meaningful.

J comes from a very small energy difference: $E(HS) - E(BS)$ is only 22–54 cm^{-1} , about $244 \times 10^{-6} E_h$, which is 7 orders of magnitude smaller than the total energy and 5 orders smaller than the relativistic energy shift. $J(1)$ and $J(3)$ are nearly equal because $\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS} \approx 1$ here (weak coupling), and $J(2)$ is half of $J(1)$ by definition, so the formula used must always be stated.

With the SVP basis sets, $J(3)$ changes from +37.8 cm^{-1} (non-relativistic) to –21.6 cm^{-1} (DKH) and –53.2 cm^{-1} (ZORA). These changes are negligible compared with the change in total energy, but they are as large as J itself and even change its sign, from weakly ferromagnetic to weakly antiferromagnetic. DKH and ZORA also differ from each other by 32 cm^{-1} . Repeating the three runs with a finer grid (DefGrid3, VeryTightSCF) changed $J(3)$ by less than 0.1 cm^{-1} , so these numbers are not grid noise. To see whether they are converged with respect to the basis set, the calculations were repeated with TZVP basis sets.

Table 2. B3LYP results with SVP-level (def2-SVP, DKH-def2-SVP, ZORA-def2-SVP) and TZVP-level (def2-TZVP, DKH-def2-TZVP, ZORA-def2-TZVP) basis sets, same geometry and BrokenSym setup.

Treatment	E(HS)–E(BS) / cm ^{–1}		J(3) / cm ^{–1}	
	SVP	TZVP	SVP	TZVP
Non-relativistic	–38.0	+78.69	+37.8	–78.11
DKH	+21.7	+127.84	–21.6	–126.87
ZORA	+53.5	+127.84	–53.2	–126.87

With the larger basis all three calculations give antiferromagnetic coupling, $J(3) = -78.1 \text{ cm}^{-1}$ (non-relativistic), -126.87 cm^{-1} (DKH) and -126.87 cm^{-1} (ZORA). DKH and ZORA now agree to 0.003 cm^{-1} , and the non-relativistic value is already negative without relativity. The sign change and the DKH/ZORA difference seen with SVP therefore come from the small basis set, not from relativity. With TZVP, scalar relativistic effects shift J by -49 cm^{-1} and make the antiferromagnetic coupling about 62 % stronger.

For comparison, changing the functional at the non-relativistic SVP level changes J(3) much more: $+173 \text{ cm}^{-1}$ with TPSSh (10% exact exchange), $+38 \text{ cm}^{-1}$ with B3LYP (20%) and -31 cm^{-1} with PBE0 (25%).

How important are the relativistic effects on J?

Relativity changes the total energy by tens of hartree, but this is core-electron stabilisation that is the same in the high-spin and broken-symmetry states and cancels in J. The effect that remains on J is about 49 cm^{-1} , or 62 % of $|J|$. For a 3d metal such as Cu this is small in absolute terms, but it is not negligible because J itself is a small energy. DKH and ZORA give the same J once the basis set is large enough, so the choice between them does not matter here. The table also shows that the SVP basis is too small to resolve the relativistic effect on such a small energy difference, and that the choice of functional affects J more than relativity does.